

Java - Threads and Concurrency

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Agenda

- Intro
- Synchronization
- ThreadPool
- Thread Local Variables
- Fork-Join (extra materials)

What are Threads?

- A thread is a lightweight sub-process, a smallest unit of processing.
- Allows multiple tasks to be executed concurrently with a single program

- Examples?

Creating a thread in Java

- Extending the Thread class
- Implementing the Runnable interface
- Using the Callable interface and Future

Creating a thread in Java - Thread class

A thread can be created by extending the Thread class and overriding the ‘run’ method.

```
class MyThread extends Thread {
    @Override
    public void run() {
        System.out.println( "Thread is running" );
    }
}

MyThread t = new MyThread();
t.start();
```

Creating a thread in Java - Thread class

A thread can be created by extending the Thread class and overriding the ‘run’ method.

```
class MyThread extends Thread {
    @Override
    public void run() {
        System.out.println( "Thread is running" );
    }
}

MyThread t = new MyThread();
t.start();
```

- Pros: Simple to implement
- Cons: Not flexible, Thread and task are coupled

Creating a thread in Java - Runnable interface

A thread can be created by implementing the Runnable interface

```
class MyRunnable implements Runnable {
    @Override
    public void run() {
        System.out.println("Runnable is running");
    }
}

Thread t = new Thread(new MyRunnable());

t.start();
```

Creating a thread in Java - Runnable interface

A thread can be created by implementing the Runnable interface

```
class MyRunnable implements Runnable {
    @Override
    public void run() {
        System.out.println("Runnable is running");
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Thread t = new Thread(new MyRunnable());

t.start();
```

Creating a thread in Java - Runnable interface

A thread can be created by implementing the Runnable interface

```
class MyRunnable implements Runnable {
    @Override
    public void run() {
        System.out.println("Runnable is running");
    }
}

Thread t = new Thread(new MyRunnable());

t.start();
```

- Pros: Solving the flexibility problem, Decouples task from thread
- Cons: A bit more complex

Creating a thread in Java - Callable & Future

Resembles Runnable, but can throw exceptions and return result

```
class MyCallable implements Callable<String> {

    public String call() {
        return "Callable result";
    }
}
```

Functional Interface

```
@FunctionalInterface
public interface Callable<V> {
    V call() throws Exception;
}
```

A functional interface is an interface with **exactly one abstract method**.

A lambda function can implement a functional interface if:

- It has matching parameters and return type
- It has compatible checked exceptions (it can also not have any checked exceptions - as is the case in regular implementing methods)

Read more:

https://docs.oracle.com/javase/8/docs/api/java/lang/FunctionalInterface.html

Creating a thread in Java - Callable & Future

```
public static void main(String[] args) {
    ExecutorService executor = Executors.newSingleThreadExecutor();
    Callable<String> myCallable = () ->  "Test";
    // non-blocking asynchronous submission of the callable to the executor
    Future<String> future = executor.submit(myCallable);
    try {
       // blocking with a timeout
        String result = future.get( 100, TimeUnit.MILLISECONDS);
        System.out.println(result);
    } catch (InterruptedException | ExecutionException e) {
        e.printStackTrace();  // Better to replace with appropriate logs
    } catch (TimeoutException e){
        // Program fallback logic + appropriate logs
    }
    executor.shutdown();
}
```

Creating a thread in Java - Callable & Future

```
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```

Creating a thread in Java - Callable & Future

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    }
    executor.shutdown();
}
```

Creating a thread in Java - Callable interface

- Pros:
  - Returns result!
  - Can throw Checked exceptions
  - In overall, a better control over task execution and result retrieval
- Cons:
  - Complex (over alternatives)
  - Requires managing helpers as `ExecutorService`

Creating a thread in Java - Summary

- Extending the Thread class - Simple but inflexible, Thread is coupled with the task
- Implementing the Runnable interface - Flexible, Decouples task from thread
- Using the Callable interface and Future - Suitable for tasks that return results/ throws exceptions, giving us the control on the execution

Creating a thread in Java - Sanity Test

```
public static void main(String[] args) {
    Thread t = new Thread(() -> System.out.println("yo!"));
    t.run();
    System.out.println("main");
}


// what will happen when running this main?
```

Thread Lifecycle

- **New** - Thread is created but not started (after new)
- **Runnable** - Thread is ready to run (after the .start())
- **Blocked** - Thread is blocked (Synchronized block, lock)
- **Waiting** - thread is waiting to another thread (Wait())
- **Timed Waiting** - Thread is waiting for a specific time (Sleep())
- **Terminating** - Thread has completed execution (noramly/ Exception)

Setting thread parameters

```
public Thread(ThreadGroup group,
              Runnable target,
              String name,
              long stackSize)
```

https://docs.oracle.com/en/java/javase/11/docs/api/java.base/java/lang/Thread.html#<init>(java.lang.ThreadGroup,java.lang.Runnable,java.lang.String,long)

ThreadGroup - ?

https://stackoverflow.com/questions/1649133/what-is-the-benefit-of-threadgroup-in-java-over-creating-separate-threads

https://wiki.sei.cmu.edu/confluence/display/java/THI01-J.+Do+not+invoke+ThreadGroup+methods

Stack size - relevant? read the documentation, validate, tune!

See also - setting thread priority:

https://docs.oracle.com/en/java/javase/11/docs/api/java.base/java/lang/Thread.html#setPriority(int)

Setting thread parameters

Setting thread priority - relevant? read the documentation, validate, tune!

```
public final void setPriority(int newPriority)
```

See: https://docs.oracle.com/en/java/javase/11/docs/api/java.base/java/lang/Thread.html#setPriority(int)

https://stackoverflow.com/questions/1617963/setting-priority-to-javas-threads

https://www.javamex.com/tutorials/threads/priority.shtml

https://www.javamex.com/tutorials/threads/priority_what.shtml

setting thread affinity

Setting thread affinity to a particular core - relevant?

need to use external library for that…e.g.

https://github.com/OpenHFT/Java-Thread-Affinity

(Note: https://github.com/peter-lawrey/Java-Thread-Affinity is not another lib, it's just an older version of the above).

See: https://stackoverflow.com/questions/2238272/java-thread-affinity

Based on JNA (Java Native Access): https://github.com/java-native-access/jna

Before Daemon Thread

```
public static void main(String[] args) {
  Thread t = new Thread( () -> {
    try{
      Thread.sleep(1000);
    }catch(InterruptedException e ){
      System.out.println("Thread Interrupted!");
    }
    System.out.println("yo!");
  });
  t.start();
  System.out.println("main");

}
// what will be printed?
```

Before Daemon Thread

```
public static void main(String[] args) {
  Thread t = new Thread( () -> {
    try{
      Thread.sleep(1000);
    }catch(InterruptedException e ){
      System.out.println("Thread Interrupted!");
    }
    System.out.println("yo!");
  });
  t.start();
  System.out.println("main");

}
// what will be printed?
```

Before Daemon Thread - nicer syntax

```
public static void main(String[] args) {
    Thread t = new Thread( () -> {
        // com.google.common.util.concurrent.Uninterruptibles
        Uninterruptibles.sleepUninterruptibly( 1, TimeUnit.SECONDS);
        System.out.println( "yo!");
        });
    t.start();
    System.out.println( "main");
}

// what will be printed now?
```

Daemon Thread

```
public static void main(String[] args) {
    Thread t = new Thread( () -> {
        // com.google.common.util.concurrent.Uninterruptibles
        Uninterruptibles.sleepUninterruptibly( 1, TimeUnit.SECONDS);
        System.out.println( "yo!");
    });
    t.setDaemon(true);
    t.start();
    // Uninterruptibles.sleepUninterruptibly(2, TimeUnit.SECONDS);
    System.out.println( "main");
}

// what will be printed now?
```

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- Synchronization
- ThreadPool
- Thread Local Variables
- Fork-Join (extra materials)

Synchronization

- Prevents memory consistency errors
- Synchronized method

```
class Counter {
   private int count = 0;
   public synchronized void increment() {
       count++;
   }
   public int getCount() {
       return count;
   }
}
```

- When a thread calls the synchronized method, it acquires the intrinsic lock. No other thread can call the method

Synchronization

A “crash course” on data corruption and ‘mutual exclusive’:

```
public class SynchronizedCounter {
  private int c = 0;
  public synchronized void increment() { c++; }
  public synchronized void decrement() { c--; }
  public synchronized int value() { return c; }
}
```

The synchronized keyword on a method means that if this is already locked anywhere (on this method or elsewhere) by another thread, we need to wait till this is unlocked before we can enter



Synchronization

- In java every object has an **intrinsic** lock (known as monitor)
- Synchronized block

```
class Counter {
   private int count = 0;
   private final Object lock = new Object();
   public void increment() {
       synchronized (lock) {
           count++;
       }
   }
   public int getCount() { return count; }
}
```

- The thread acquires a specific lock (lock)
- Synchronization is only for the critical section

Synchronization

```
public class TwoCounters {
    private long c1 = 0, c2 = 0;
    private Object lock1 = new Object();
    private Object lock2 = new Object();
    public void inc1() {
        synchronized(lock1) {
            c1++;
        }
    }
    public void inc2() {
        synchronized(lock2) {
            c2++;
        }
    }
}
```

Synchronization

```
public void addName(String name) {
  synchronized(this) {
      lastName = name;
      nameCount++;
      nameList.add(name);
  }
  // do things that don't require synchronization
}
```

When synchronizing a block, key for the locking should be supplied (usually would be this).

The advantage of not synchronizing the entire method is efficiency

Synchronization

What does synchronize on a static method mean?

```
public class SynchronizedCounter {
  public static synchronized int getValue() {
  // do something that need to be synchronized
  }
}
```

The lock is the class object representing this class.

Note: synchronized on static method should be very careful.

Atomic Variables

Allows atomic operations on a variable without synchronization.
Usually should be more efficient.

Implementation is Hardware and OS dependent but may use Compare and Swap (CAS).

See: https://docs.oracle.com/javase/tutorial/essential/concurrency/atomicvars.html
https://docs.oracle.com/javase/9/docs/api/java/util/concurrent/atomic/package-summary.html

Atomic Variables

```
import java.util.concurrent.atomic.AtomicInteger;

public class AtomicCounter {
  private AtomicInteger c = new AtomicInteger(0);
  public void increment() { c.incrementAndGet(); } //No synchronization
  public void decrement() { c.decrementAndGet(); }
  public int value() { return c.get(); }
}
```

Additional locks and Sync Utils

java.util.concurrent.locks

- LockSupport park, unpark, parkUntil
- ReentrantLock + Condition
  (resembles wait notify, with: await, signal, see:
  https://stackoverflow.com/questions/10395571/condition-vs-wait-notify-mechanism)
- ReentrantReadWriteLock vs. StampedLock -- both for Read-Write
  (see: https://stackoverflow.com/questions/28918004/readwritelock-vs-stampedlock)

java.util.concurrent

- Semaphore - allow N acquires before blocking
- CountDownLatch, CyclicBarrier, Phaser - wait for completion of N activities
  (see: https://stackoverflow.com/questions/184147/countdownlatch-vs-semaphore)
- Exchanger - exchange objects between two threads in a thread safe way

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Thread Pools - Executor Service

- Managing a pool of worker threads and provides methods to submit tasks for execution.
- Simplifies the process of managing threads - in compare to manual thread creation.
- Efficient reuse of threads, reducing creation time/ destruction time.
- Java interface in the java.util.concurrent package.

Avoid using raw thread!!! Use executor and thread pools.

Creating Thread Pools

- Fixed Thread Pool
  - Fixed # of thread

```
ExecutorService fixedThreadPool =
Executors.newFixedThreadPool( 4);
```

- Cached Thread Pool
  - Creates new thread as needed
  - Uses previously constructed threads

```
ExecutorService cachedThreadPool =
Executors.newCachedThreadPool();
```

Creating Thread Pools - cont.

- Single Thread Executor
  - A ‘pool’ with one thread
  - Ensures that tasks are executed sequentially in the submitted order

```
ExecutorService singleThreadExecutor =
Executors.newSingleThreadExecutor();
```

- Schedule Thread Pool
  - Can schedule delay tasks / execute periodically tasks

```
ScheduledExecutorService scheduledThreadPool =
Executors.newScheduledThreadPool(2);
```

Example - Fixed Thread Pool

```
public class FixedThreadPoolExample {
    public static void main(String[] args) {
        ExecutorService executorService = Executors.newFixedThreadPool( 4);
        for (int i = 0; i < 10; i++) {
            final int taskId = i;
            executorService.submit(() -> {
                System.out.println( "Task "+taskId+"is running thread "
                                    +Thread.currentThread().getName());
                try { Thread.sleep(2000); } // Simulate thread work
                catch (InterruptedException e) {
                   Thread.currentThread().interrupt();
                }
                System.out.println( "Task " + taskId + " completed");
            });
        }
        executorService.shutdown();
    }
}
```

Exercise - ThreadPool

Read from a file line by line and send each line as a task to a ThreadPool.

Each line should have userID and userName in the following format:

userid=<userID>, username=<userName>

Example input:

userid=123, username=Momo

Create a Map of userID to userName.

Optional (When the base functionality works):

1. in case where a userID already exists:
   a. If the name is the same - ignore.
   b. If the name is different, keep the one that is already in the map and log the event.
2. Lines that do not follow the format should be copied to an Errors collection (of lines + appropriate logs).

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Thread Local Variables

- `ThreadLocal` provides variables that each thread accessing them has its own independent initialized copy.
- Useful for Storing per-thread context information -
  - User session management
  - Database connection management
- No need for explicit synchronization when accessing (reducing the complexity of the concurrent code)

Thread Local Variables - Toy example

```
public class ThreadId {
    // Atomic integer containing the next thread ID to be assigned
    private static final AtomicInteger nextId = new AtomicInteger(0);

    // Thread local variable containing each thread's ID
    private static final ThreadLocal<Integer> threadId =
        new ThreadLocal<Integer>() {
            @Override
            protected Integer initialValue() {
                return nextId.getAndIncrement();
            }
        };

    // Returns the current thread's unique ID, assigning it if necessary
    public static int get() {
        return threadId.get();
    }
}
```

Thread Local Variables - Toy example

```
public class ThreadId {
    // Atomic integer containing the next thread ID to be assigned
    private static final AtomicInteger nextId = new AtomicInteger(0);

    // Thread local variable containing each thread's ID
    private static final ThreadLocal<Integer> threadId =
        new ThreadLocal<Integer>() {
            @Override
            protected Integer initialValue() {
                return nextId.getAndIncrement();
            }
        };

    // Returns the current thread's unique ID, assigning it if necessary
    public static int get() {
        return threadId.get();
    }
}
```

Anonymous subclass!

Thread Local Variables - Simpler syntax

```
public class ThreadId {
    // Atomic integer containing the next thread ID to be assigned
    private static final AtomicInteger nextId = new AtomicInteger(0);

    // Thread local variable containing each thread's ID
    private static final ThreadLocal<Integer> threadId =
        ThreadLocal.withInitial( ()  -> nextId.getAndIncrement() );

    // Returns the current thread's unique ID, assigning it if necessary
    public static int get() {
        return threadId.get();
    }
}
```

User session management - complex Example

Let’s design a user session management

- Using Thread Local
- Multiple sessions - for the example 2 :) in the system.
- Session type - string
- Support - get, set, and remove

User session management

```
public class UserSession {
    private static final ThreadLocal<String> userSession =
                                              ThreadLocal.withInitial(() -> null);
    public static void set(String user) {
        userSession.set(user);
    }

    public static String get() {
        return userSession.get();
    }

    public static void remove() {
        userSession.remove();
    }
}
```

User session management

```
public static void main(String[] args) {
  Runnable session1 = () -> {
        UserSession.set("User1");
        System.out.println("session1: "
                  + UserSession.get());
        UserSession.remove();
            };

  Runnable session2 = () -> {
        UserSession.set("User2");
        System.out.println("session2: "
                  + UserSession.get());
        UserSession.remove();
    };
  //cont
```

```
//cont
      Thread user1 = new
      Thread(session1);
      Thread user2 = new
      Thread(session2);

      user1.start();
      user2.start();
}
```

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Fork/Join Framework

- Designed to efficiently handle tasks that can be broken down into smaller subtasks and process parallel
- Key Components
  - ForkJoinPool (implementation of ExecutorService)
  - RecursiveTask<V> - Fork tasks and join their results
  - RecursiveAction - Fork tasks without join the results

    (Both RecursiveTask & RecursiveAction need to override compute() method)

Fork/Join Framework - Fibonacci Example

```
class FibonacciTask extends RecursiveTask<Integer> {
   final int n;
   FibonacciTask(int n) {
        this.n = n;
   }

   @Override
   protected Integer compute() {
       if (n <= 1) return n;
       FibonacciTask recFib1 = new FibonacciTask(n - 1);
       recFib1.fork();
       FibonacciTask recFib2 = new FibonacciTask(n - 2);
       return recFib2.compute() + recFib1.join(); //Join the forked result
   }
}

ForkJoinPool pool = new ForkJoinPool();
int result = pool.invoke(new FibonacciTask(10));
System.out.println(result);
```

That concludes this chapter

Thank You